McLean
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ENERGYWAVE, INC.

Energy at the Point of Use

The Power Bottleneck in Ambulatory Robotics, and a Structured Research Program to Address It

Prepared for Department of Defense technical review
EnergyWave, Inc. (Delaware, in formation) — a McLean Companies venture
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Statement of Claims Discipline

This document makes no performance claims about any device.

EnergyWave has no prototype, no measured output, and no validated energy conversion mechanism. Nothing in this paper should be read as an assertion that such a mechanism exists or has been demonstrated.

What this document does assert is narrower and, we believe, defensible:

1. Portable energy density is the binding constraint on ambulatory robotic platforms, and the constraint is worsening rather than easing.
2. Incremental electrochemical improvement will not close the gap on the timeline that autonomy capability is arriving.
3. A small, disciplined research program with public retirement criteria is a low-cost, high-information way to test whether any non-chemical approach merits further national attention — and to close the question publicly if it does not.

We state this at the front because the field we are entering has a long history of unfalsifiable claims. We intend to be legible as the opposite of that.

1. The Bottleneck

Progress in information technology has compounded for five decades. Progress in portable energy density has not. Transistor counts have risen by roughly seven orders of magnitude since 1971. Practical specific energy in portable storage has risen by less than one.

Every autonomous physical system inherits that divergence. The intelligence is arriving; the power to move it is not.

1.1 Current platform reality

The commercial humanoid platforms now entering production illustrate the constraint precisely.

Tesla's Optimus carries a 2.3 kWh lithium-ion pack at roughly 52V. Manufacturer framing describes approximately eight hours of light-duty runtime; under continuous heavy load such as lifting or fast walking, reported real-world runtime is closer to three to six hours. The next-generation platform does not

increase pack capacity — battery is unchanged at approximately 2.3 kWh, with improvement targeted through efficiency rather than a larger pack. [1][2]

Across the sector the picture is consistent. Runtime for leading humanoid platforms ranges from ninety minutes to eight hours per charge depending on task intensity, with swappable packs used to enable continuous shift coverage. Commercial platforms typically run two to five hours per charge under working load; Agility Digit and Unitree H1 fall closer to one and a half to two hours in high-activity scenarios. [3]

The gap between specification and field performance is itself now a recognized problem. Published runtimes are derived from controlled environments and calculated against favorable duty cycles, while battery capacity and runtime remain the primary bottlenecks to commercial viability. [4]

1.2 Why actuation is the hard case

The load is not computation. It is mechanical work.

Actuators account for 56% of the robot's total bill-of-materials cost on a production humanoid, and a current-generation platform carries 28 body actuators plus 50 hand actuators across 22 degrees of freedom. [2] Every degree of freedom added for dexterity is a continuous power draw under gravity load, not an intermittent one. A robot holding a position consumes power. A robot walking consumes substantially more.

This is the structural difference between robotics and every prior computing platform. Moore's Law made the thinking cheap. It did nothing for the lifting.

1.3 Why chemistry will not close it

Lithium-ion is approaching its material ceiling. Theoretical energy density for lithium-ion ranges from 400–500 Wh/kg against a current practical range of 150–250 Wh/kg, and the practical figure is constrained by thermal management, packaging, and safety margin rather than by cell chemistry alone. [5] Industry planning assumptions put improvement at roughly 4% per year. [6]

At 4% compounding, doubling takes approximately eighteen years. Platform manufacturers are not waiting for it — they are engineering around it. Tesla has filed for an autonomous charging station design; Figure 03 integrates charging coils into the soles of its feet, stepping onto an inductive pad that delivers 2 kW. Manufacturers have largely abandoned lithium iron phosphate in favor of high-nickel ternary chemistries to obtain the energy density and high-rate discharge required to power dozens of simultaneous joint movements — and have accepted the cost and thermal penalties to do it. [7]

These are competent responses to a hard constraint. They are also an admission of it. The industry's current answer to endurance is logistics — swap packs, dock, run in shifts — not energy.

2. Why This Is a Defense Problem

The commercial framing above understates the defense relevance, because the commercial case tolerates a charging dock and the defense case often does not.

The dismounted power burden is a decades-old, unsolved logistics problem. A dismounted infantryman or scout in Afghanistan carried roughly 9.7 pounds of batteries, unable to recharge when not near a vehicle or

a forward operating base; battery weight was projected to exceed 14 pounds for a 72-hour mission as every soldier is brought into the network. [8] Allied assessments place the burden as high as 25% of a soldier's carried load on some mission profiles. [9]

The Army has stated the trend plainly: across all six modernization priorities there has been a dramatic increase in demand for power as a result of energy-hungry new system capabilities, with particular concern in soldier lethality including augmented reality. [10]

Now add an ambulatory robotic platform to that formation. A squad-accompanying system at current energy density either tethers the unit to a resupply and recharge cycle, or it does not deploy. Endurance becomes a doctrinal constraint, not an engineering footnote.

The strategic question follows directly: **a five-fold improvement in portable energy density at the point of use changes force structure. A 4%-per-year improvement does not.** That gap is worth a small, well-governed research investment even at low probability of success, because the payoff is asymmetric and the cost of being second is high.

3. What We Propose

EnergyWave proposes a research program, not a product program.

3.1 The protocol

Every candidate mechanism we examine proceeds through three gates, in order, with no gate skipped:

Characterize. Establish precisely what the claimed effect is, under what conditions, with what instrumentation, and what the null hypothesis predicts. Reproduce the original measurement setup where records permit.

Conservation-test. Subject the effect to closed-system energy accounting. Every input path is measured — including ambient thermal, RF, magnetic, and mechanical sources. If the observed output is accounted for by a conventional input, the mechanism is reclassified as a transducer, not a source, and evaluated on those honest terms.

Extract or retire publicly. If an effect survives conservation-testing, proceed to engineering characterization. If it does not, we publish the negative result under our own name.

The third gate is the one that distinguishes this program. Negative results in this field are almost never published, which is precisely why the same claims recirculate for decades and why serious institutions will not engage with the area. We will publish ours. A retired claim is a deliverable.

3.2 What we are not claiming

We are aware that the physics literature bearing on vacuum energy extraction is not favorable. Peer-reviewed reviews of quantum vacuum energy extraction state directly that continuous cyclic extraction of this kind would violate the second law of thermodynamics, and that power figures appearing in that literature are unbuilt theoretical estimates rather than measured results.

Our own advisors have been consistent on this point. Our principal theoretical physicist has assessed at least one historically prominent solid-state approach as non-viable as described. Our senior engineering

advisor, whose career runs from MIT research through commercialization of cryogenic systems for MRI and superconducting electronics, has held the position that the second law is not incomplete.

We are not asking the Department to accept a claim. We are proposing to apply competent instrumentation and honest accounting to a set of specific historical claims, and to publish what we find in either direction.

4. Control and Safety Architecture

If any high-density portable energy source becomes practical, its control architecture is not a downstream concern. It is the enabling condition for release. We are designing to that assumption from day one.

4.1 Design principles

Platform binding. The module is scoped to ambulatory robotic platforms and cryptographically bound to a specific host. It authenticates to the platform controller and delivers no output to an unrecognized host. The functional analogy is a keyed consumable: the cartridge and the printer are useless to each other absent mutual authentication.

Application scoping by design, not by policy. Output profile — power ceiling, ramp rate, duty envelope — is fixed in hardware to the actuation and compute envelope of an ambulatory platform. The module is not general-purpose. It cannot be repurposed as a bulk supply without defeating hardware limits, which is a detectable and destructive act.

Tamper-responsive inertion. The enclosure carries a tamper-response boundary of the kind already standard in high-assurance cryptographic hardware. Envelope breach, unauthorized disassembly, out-of-spec thermal or electrical conditions, or authentication failure trigger irreversible transition to an inert, non-functional state. There is no service mode, no recovery path, and no field-repairable configuration. The module fails permanently closed.

Custody and revocation. Units are individually identified, attested, and remotely revocable. A recovered or diverted unit is a dead unit.

4.2 Negative scope

We state exclusions explicitly and intend them to be binding conditions of any government relationship:

- No directed-energy application.
- No propulsion application.
- No munitions or ordnance integration.
- No general-purpose or grid-scale power configuration.
- No export or foreign transfer absent explicit U.S. Government authorization.

We would welcome these being written into an agreement rather than merely stated in a paper.

4.3 Disclosure posture

We are structuring toward controlled rather than open disclosure. The reasoning is simple: an energy density improvement of the magnitude that would matter is dual-use by construction, and the appropriate time to

establish government visibility is before results exist, not after. We would rather be inside a framework early than negotiate one under pressure later.

5. Program Structure

Phase 0 — Formation and Instrumentation (0–6 months). Delaware C-corp formation. IP position established through counsel. Laboratory instrumentation specified and calibrated to conservation-test standard. Independent measurement partner identified. Advisory board seated. *Gate: instrumentation validated against known standards before any candidate mechanism is energized.*

Phase 1 — Characterization (6–18 months). Candidate mechanisms characterized and conservation-tested in priority order. Each candidate exits with a published determination. *Gate: at least one mechanism survives conservation-testing with independent witness, or the program publishes comprehensive negative findings and the venture winds down or repurposes.*

Phase 2 — Engineering Characterization (18–36 months). Contingent entirely on Phase 1 outcome. Scale, stability, thermal behavior, and manufacturability assessed. Control architecture prototyped and adversarially tested independent of any energy result. *Gate: reproducible bench performance under third-party observation.*

Phase 3 — Integration (36+ months). Contingent on Phase 2. Platform integration under Government safety and security framework.

The control and safety architecture in Section 4 is developed on its own track and does not depend on Phase 1 succeeding. It has standalone value for any high-density portable energy source, including conventional ones.

6. Team

Ian Franklin McLean — Founder. M.S. Computer Science, UC Irvine, with quantum computing coursework. Prior: Wolfram Research; venture technology at Avnet with early engagements including Anduril and Wrap Technologies (NASDAQ: WRAP); FDIC during the 2008 financial crisis. Founder of Statpedia. Prior NDA consulting engagement with a U.S. intelligence organization.

Jack Sarfatti, Ph.D. — Principal, Theoretical Physics. Cornell, under Hans Bethe. Member of the Fundamental Fysiks Group, credited in the historical lineage of quantum information science.

Alan Crunkleton, Ph.D., M.D. — Senior Technical Advisor. Ph.D. Mechanical Engineering, MIT; M.D., University of Massachusetts. Spun MIT research into a cryogenics company serving MRI and superconducting electronics markets; CEO approximately eight years through acquisition. Partner, Hardesty LLC. 35+ years concept to commercialization.

Mike Wentz — Technical Diligence. MedFocus, Lake Forest, California. University of Pittsburgh. Technology leader partnering with medical device startups.

Chip Story — Physicist. R&T Team Leader, ZPEPG, a 501(c)(3) research organization.

Legal. K&L Gates — Brian Novak, Partner, Orange County; Dennis Majewski, Partner, holding an advanced degree in electrical and computer engineering with prior experience as a senior electrical design engineer in semiconductors, quantum circuits, and micro sensors.

7. The Ask

We are requesting a technical review meeting to discuss:

4. Whether the ambulatory platform power constraint described in Sections 1–2 is recognized as a capability gap within relevant program offices.
5. Whether a conservation-tested characterization program with public retirement criteria is of interest as a low-cost, high-information investment.
6. Whether the control architecture in Section 4 should be developed under Government specification, independent of the energy research outcome.
7. What framework would govern disclosure, custody, and scope restriction should Phase 1 produce a positive result.

We are not asking for validation of a claim. We are asking whether this question is worth answering carefully, by people willing to publish a negative.

Sources

[1] Tesla Optimus battery life and runtime specifications, 2026. optimusk.blog

[2] Tesla Optimus hardware: actuators, hands and sensors, 2026. optimusk.blog

[3] How humanoid robots work: complete 2026 guide. theresarobotforthat.com

[4] Humanoid robot battery reality: spec sheets vs. real-world deployment. RobotWale, April 2026.

[5] Exploring the theoretical energy limits of lithium-ion batteries, 2025.

[6] Energy density improvement projections, U.S. Patent 11542032 (hybrid-electric aircraft energy allocation).

[7] Battery technology and the 24/7 robot challenge. nexthumanoid.com

[8] Meredith, "Soldier Power: A Growing Operational Concern," Infantry Magazine, U.S. Army, Jul–Sep 2013.

[9] "Portable power storage for the dismounted soldier," European Security & Defence, February 2024.

[10] "Army researchers hope to lighten Soldiers' battery load," U.S. Army, August 2018.

[11] National Research Council, Energy-Efficient Technologies for the Dismounted Soldier; Meeting the Energy Needs of Future Warriors.

This document contains no classified information and no export-controlled technical data. It describes no device design, fabrication method, or operating parameter.

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